

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

This appendix sets out the three things every threshold quoted in this paper is checked from: the frame of each frequency, the instrumental response, and the primary-beam factor that turns a measured noise into a flux at the star.

A.1. Frequency frames, linewidth and smearing

Every frequency axis here — searched spectra, tabulated ranges and crossing frequencies alike — is *topocentric* as delivered by the ALMA correlator, and the search itself applies no velocity-frame shift. Velocities enter at the vetting stage, where the line mask is evaluated in each star’s own rest frame: it carries every crossing frequency to the barycentre at its own block’s epoch and then to the stellar frame before comparing it with a transition (Appendix C). The residual error is then the catalogue radial-velocity uncertainty alone, a few km s^{-1} within 40 pc; second-order relativistic terms at the 50 km s^{-1} scale are $\lesssim 5 \text{ kHz}$ at 350 GHz, below the finest native channel searched. The stacking is drift-aligned and inverse-variance weighted: coherent in frequency trajectory, not in electric-field phase.

The native channel width is also the linewidth convention for every EIRP threshold. The statistic is the single-channel flux density after drift correction, so $\text{EIRP}_{\text{trig}}$ is the minimum *total* power of a channel-confined carrier, flat in the assumed linewidth below the channel and never a spectral power density; flatness is exact only for a centred tone, and a boundary-straddling one divides its power between two channels, which is the loss the injections carry and one reason EIRP values are quoted to two significant figures. The Hz-resolution surveys’ limits are total powers in the same sense, each the best case of its own channelisation, which makes the axes of Fig. 1(a) commensurable. One number fixes the scale, illustratively and not achievably: rechannelising the median Class A window’s 488 kHz to the 2.79 Hz of a dedicated narrowband search would divide the threshold by 418, and no reprocessing recovers resolution the correlator never recorded.

Channel widths come from the measurement sets’ own `CHAN_WIDTH`; MS v2 defines no “EFFECTIVE_BANDWIDTH” column, so the quoted channelisations are separations and are not claimed to be noise bandwidths.

ALMA applies online Hanning smoothing by default (Cortes et al. 2026), so the spectral response is ~ 2 channels wide and neighbouring channels are correlated. For the 0.25/0.5/0.25 kernel the coefficients are $\rho_1 = 2/3 = 0.667$ and $\rho_2 = 0.167$, with $\sigma_{\text{smoothed}}/\sigma_{\text{raw}} = 0.612$: the correlation is not confined to adjacent channels, the lag-two term being a sixth. The peak channel holds 0.50 of an unresolved carrier’s flux at a channel centre and 0.42 on a boundary, a median 0.48 over sub-channel phase, so a nominal threshold is optimistic by $\times 2.00$ – $\times 2.36$, and every threshold quoted here carries that correction window by window.

The injected carriers are deposited with the correlator’s own channel response, at a sub-channel phase drawn afresh for each tone, so the smoothing loss sits inside the measured completeness rather than being applied to it. The deposition falls 8 per cent short of the ideal response averaged over phase, so the completeness is conservative by that much. Where in a channel a carrier was placed makes no difference to its recovered signal-to-noise, because a carrier at a typical searched drift rate crosses about 11 channels during one observation and so samples the whole response. Since the loss is carried end to end it must not be applied to the result as well: at the ninety per cent point of the recovery curve the peak channel holds $1.93\times$ the nominal trigger flux, which is that factor of about two, already spent.

Two measurements fix the smoothing state. The retained per-integration residuals of the second-stage local null have a lag-one spectral autocorrelation of 0.65 against the Hanning prediction 0.667, so the correlation is instrumental and those spectra are smoothed and un-averaged; and the archive’s reported effective spectral resolution is exactly twice the channel separation in 1505 of the 1651 windows, in both correlator modes; the other 146 report a smaller ratio, the signature of online channel averaging, where the peak-channel loss is smaller. Every window has a reported resolution, so the blanket factor is right for most of both classes. *The correlation model is truncated at lag two*: the baseline step, a per-integration median over up to 65 channels, could imprint longer-range correlation that the retained spectra carry no residuals to measure, which would make the independent-trials count of Appendix D an over-estimate.

Transmitters wider than one channel. The baseline filter (§4.1) is flat for features narrower than roughly half its effective width W and falls to zero above W , so the survey’s sensitivity is to channel-confined morphologies and $\bar{W}$ is the linewidth ceiling: 65–74 channels over the fine class and 30–59 over the coarse one, that is $\approx 32 \text{ MHz}$

¹ School of Physical Sciences, The Open University, Walton Hall, Milton Keynes, MK7 6AA, UK; glenn.white@open.ac.uk

² RAL Space, STFC Rutherford Appleton Laboratory, Harwell Campus, Didcot, OX11 0QX, UK

³ VBRL Holdings, 50/5 Huay Kaew Road, Chang Phueak, Chiang Mai, Thailand

at 488 kHz and ≈ 0.9 GHz at 15.62 MHz. Within that ceiling a top-hat transmitter spanning N_{ch} native channels has per-channel flux the total divided by N_{ch} at unchanged noise, so its threshold is N_{ch} times the single-channel one.

Intra-integration smearing, which the injections do not model. Each injected tone is deposited at one frequency per integration, so the campaign carries the channel response and the drift between integrations but not the sweep within one. A carrier moving Δ channels during an integration keeps $\eta_{\text{smear}} = \text{sinc}(\pi\Delta/2)$ of its peak amplitude, and every limit in this paper is divided by it. Since EIRP₉₀ is a ninety per cent point over carriers drawn from each window's own drift grid, the Δ that sets it is the one at the ninetieth percentile of that draw, 89 per cent of the searched ceiling over the campaign's 10 332 tones. The integration length is recorded for 1046 of the 1651 windows, 451 of them in the delivered meta-data and 595 more in the exposure records the search kept; for the other 605 the factor is *bounded* at 11.6 s, the longest integration delivered anywhere, where none falls below 0.997. Nothing moves in the median window, and 13 windows on 6 stars rise by more than one per cent, up to $\times 1.53$: HIP 10679, HD 172555, β Pic, η Crv, CP-72 2713 and HD 48370. Re-scoring each of the 49 injected windows' own recovery ladders with every tone de-rated by its own η_{smear} gives a penalty never larger than the one applied and smaller by at most 8 per cent, so the correction errs the safe way.

A.2. The primary-beam response, and the two columns that let a reader recompute it

The sensitivity quoted for a window is $S_{\text{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre: the noise σ is measured in apparent flux, the limit is quoted in true flux, and the two differ by A . That offset is not a nuisance. Archival fields are pointed at a catalogue position, often decades old, while these are high-proper-motion stars, so the star can sit well off axis: over the 1651 windows the offset has a median of $0.089''$, a 99th percentile of $7.43''$ and a maximum of $8.13''$, and A runs from 0.999984 in the median down to 0.5522, below 0.99 on 91 windows and below 0.9 on 34. *None* falls below one half, and every one of the low-response windows is a physically explicable J2000 pointing on a fast-moving star rather than an error. A window is kept only where the response is at least 0.5, a correction of at most $\times 2$, since at low response a small error in the beam model becomes a large error in inferred flux. The offsets are bimodal and the floor sits in the gap: the last retained window has response 0.55 at $0.46\theta_{\text{PB}}$, the first window beyond it 0.35 at $0.62\theta_{\text{PB}}$, and nothing lies between, so any floor between those two responses retains the same 1651 windows and the value chosen does not matter. Beyond it lie the 12 ϵ Eridani Band 6 windows, which were withheld as a band and not window by window. 8 of them, in 2 blocks, need a correction of up to $\times 3.5$ — past the first sidelobe transition, where the Gaussian form fails — and so fail the floor outright. The other 4, in block Xb9c01c_X2004, place the star at the pointing centre and need no correction at all, so the floor is not what removed them; and at 3.22 pc the star is too near to be left without a limit, so theirs are given here. They carry 0.10 window-hours

and no threshold crossing, the largest statistic being 3.8 where the trigger is five. The one fine-channel window, at 488 kHz, reaches 1.8×10^{14} W and the 3 coarse ones at 15.6 MHz 9.3×10^{14} – 1.1×10^{15} W; none enters a census statistic. The flux-scale comparison of §4 is a separate continuum measurement on the same star: a block mean over the 4 compact-array executions that place it at the pointing centre, 0.22 h in Band 6, where no beam correction enters either.

The correction is applied exactly once. The offset is computed for every measurement set from its own FIELD::PHASE_DIR and the propagated stellar position, so no window carries a zero-by-assumption offset, and it enters in one place only: $S_{\text{min}} = 5\sigma/A$ closes on all 1651 catalogued rows to $8.7\text{e-}05$ in relative terms. The count is in two parts, because on 10 rows whose products are not retained A is back-solved from $5\sigma/S_{\text{min}}$ and flagged as such, so there the closure is arithmetic; the audit is a test on the other 1641 rows, where A comes from the search's own stored geometry and closes to $8.7\text{e-}05$. Nothing else needs correcting: the statistic of Eq. 1, the control ranks and the attribution are *ratios* formed inside one window, in which A divides out identically.

The response applied is a Gaussian of full width $1.22\lambda/D$ — an Airy first-null coefficient — at each window's *own* dish diameter, reproduced over all 1651 rows to 2.6×10^{-5} against 7.7×10^{-2} for the same Gaussian forced to $D = 12$ m. The distinction is not cosmetic: the 1054 rows taken on the $1.7\times$ smaller compact-array dishes have a correspondingly wider beam, and their response cannot be obtained from a 12 m beamwidth at all.

The beamwidth convention is a bounded systematic, and both conventions were computed. ALMA's primary-beam full width is $1.13\lambda/D$, narrower than the first-null coefficient, so it attenuates more and makes every limit *shallower*: by nothing at all in the median window, by more than one per cent in 40 of 1651 and by more than ten in 4, the largest being $+10.3$ per cent — those few all LP 736-15 at $8.1''$ off axis. The lowest response in the sample moves 0.5522 to 0.5005, still clear of the 0.5 retention floor but by a margin worth saying. No disposition changes with the convention. The remaining model uncertainty is also bounded by measurement: CASA's blocked-Airy response is better physics than either Gaussian, the largest ratio to it over the whole sample is 1.0745, and the counts of systems reaching 10^{15} W are identical under any of the three.

The star's apparent direction includes the annual parallax, and the amplitude a compact source retains when that is left out of the assumed phase centre is measured rather than modelled. The displacement decides whether the loss can be treated as a scalar at all: a small fraction of a synthesised beam attenuates a compact source and does nothing else, while a displacement approaching a beam puts the phase wrong on the longest baselines. Re-extracting the 8 windows of 2 blocks furthest out at the full apparent position — proper motion, parallax and radial velocity where catalogued — leaves each window's noise unchanged to 0.20 per cent and alters no disposition. It cannot test the amplitude factor, because that factor describes a source and those windows hold none: the statistic is a maximum over a plane of drift trajectories, so moving the extraction point by two thirds of a

beam draws a fresh noise realisation, and the ratio scatters from 0.78 to 1.15 about a median of 0.98 instead of approaching the 0.49 a source would give. A carrier injected at the apparent position and recovered from the *same* visibilities at both positions does return the normalised dirty beam at the omitted displacement, measured on that block’s own baselines with no model in it: 0.92 against a predicted 0.89 at 0.25 of a beam, and 0.58 against 0.49 at 0.67. That it exceeds the model at the larger displacement is the search’s doing, since maximising over a plane of trajectories lets a partly decohered carrier still find a cell, so a limit corrected by the model is shallower than it needs to be.

The injection campaign is blind to A by construction, and the bias that follows is exactly zero. The injector deposits an unattenuated amplitude at the star’s own direction cosines and refers the ladder to the *apparent* 5σ on all 40 scored units, so an injected apparent amplitude recovered nine times in ten implies a true flux larger by $1/A$ while P_{trig} is the power of a true $5\sigma/A$ and A cancels identically; 4 of the units have apparent and corrected S_{min} differing by more than a part in a thousand, so the cancellation is tested rather than vacuous. Being self-consistent in apparent flux, the campaign cannot falsify the beam model; and it reaches only $A \geq 0.9102$, whereas 75 windows on 5 stars lie further off axis, down to $A = 0.552$, so a limit quoted for one of those rests on an extrapolated curve; their 6 crossings are ratios and unaffected.

A few rows carry a catalogued noise that no retained product reproduces, and they are declared rather than reconded: on 4 windows of HD 285968 in block A002_Xd98580_X3bf6 the catalogued σ is 0.88–1.32 per cent *higher* than the surviving product’s, so the limits there are weaker and not stronger. With the 10 windows whose products are not retained the declared provenance gap is 14 windows; the attenuation columns are unaffected.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic, no disposition rests on it, and this appendix is the measurement that settles why: The star-versus-control rank is an empirically useful way of deciding which crossing to examine first, but the stellar and control positions are not demonstrably exchangeable under the null. The rank therefore has no calibrated per-window false-alarm probability and is not used as evidence for detection.

The star lies at the phase centre, inside the control annulus, and the asymmetry is real. *It is not a primary-beam effect.* Spectra are extracted from phase-rotated visibilities in apparent flux and are never corrected for the primary-beam response (Appendix A), so the familiar rise of image noise with distance from the pointing centre cannot operate here; and were it operating, it would make the control probes noisier than the star and place the stellar rank *above* one half, the opposite of what is measured. No mechanism for the displacement is established, and the one that would predict its sign is excluded below. What the control ensemble does show is a weak rise of its own level with radius, and the obvious repair is

to remove it: each probe is standardised within its own radius bin, the radii being known from the fixed draw, with no free parameter.

Applied out of sample, the correction does not repair the statistic. Frozen — probe radii, bin edges and standardisation alike — and applied unmodified to 778 windows that played no part in its construction, it moves the median stellar rank from 0.409 to 0.407 against the exchangeable one half, further from it, in both out-of-sample partitions separately. Rescoring the whole survey with the gradient removed moves the list of crossings out-ranking all their controls from 12 windows to 11, losing 2 and gaining 1, and leaves the survey median rank slightly further from one half than before.

The radial gradient is real, but removing it leaves the median no closer to one half, so the residual is not radial and neither normalisation yields a calibrated tail probability. We therefore keep the frozen rank, read it only as an indication of which crossing to examine first, and rest every disposition on physical evidence or on recurrence. Because the rank gates nothing, none of this enters a sensitivity limit.

A calibration error that varied with frequency would explain the displacement, and the usual argument does not exclude one. A gain error that is the same in every channel is removed exactly by the per-integration median along frequency; a channel-dependent one is not, and two ordinary errors are channel-dependent — a residual in the bandpass, and a system temperature measured in the coarse correlator mode and interpolated onto the science channels, which near a telluric line leaves a narrow feature behind (Hunter et al. 2018). Both multiply the continuum, so both are present at the star and absent on empty sky, both stand still in topocentric frequency, and both are coherent between integrations: they would stack at near-zero drift and put the stellar rank *below* one half, which is the sign observed.

They are excluded by the continuum these stars actually have. Each star’s $2.2\mu\text{m}$ flux extrapolated to each window as a blackbody at the star’s own $V - K$ colour temperature gives a photospheric continuum spanning 5.6 decades over the 1804 windows of the census and the hold-out whose stars have both colours, and nowhere more than 4.56 times the single-channel noise (median 0.029). A fractional error ϵ deposits ϵS_c at the star, so a crossing at the trigger needs $\epsilon = 5\sigma/S_c$: 1.10 at the most favourable window of either sample and 173 at the median, against a nominal bandpass accuracy of 0.2 per cent (Kameno 2014) and the 5–20 per cent a coarse system temperature leaves at an ozone line. **No window of this survey would admit the mechanism even were the error as large as the continuum it multiplies**, the shortfall being a factor of 5.5 at best and 865 typically. Nor does the rank follow the continuum-to-noise ratio the error would scale with: $r = +0.10$ (–0.14 to +0.32) over 76 census stars and –0.17 (–0.46 to +0.17) over 33 hold-out stars, with $T_\star$ equally flat over the 50 crossings. Against continuum *flux* the hold-out does lean, $r = -0.45$, but its continuum-brightest stars are also its most deeply observed (–0.31 against the noise alone). The same arithmetic settles the other question the shared noise scale raises: over the 1595 windows carrying no crossing the stellar statistic divided by the median of the 512 control statistics — which *is* the ratio of

TABLE A1

THE SURVEY LEDGER, PART ONE: EXECUTION BLOCKS AND WINDOWS. EACH ROW CARRIES ITS OWN NUMBER, AND A GROUP SUMS TO THE ROW WHOSE NUMBER HEADS IT — ROWS 3.1.1 TO 3.1.4 TO ROW 3.1, ROWS 3.1 TO 3.5 TO ROW 3. THE PRIMARY CENSUS IS THE SURVEY; UNLESS A STATEMENT NAMES ONE OF THE OTHER SETS, IT IS COMPUTED OVER THAT ROW. HOURS ARE WINDOW-HOURS: ON-SOURCE TIME SUMMED OVER SPECTRAL WINDOWS, SO THE CLASSES PARTITION THE HOURS AS THEY PARTITION THE WINDOWS. THE TELESCOPE ON-SOURCE TIME BEHIND THE PRIMARY CENSUS IS 280 h, OF WHICH 152 h CARRIES AT LEAST ONE CLASS A WINDOW AND 276 h AT LEAST ONE CLASS B WINDOW — FIGURES THAT OVERLAP AND MUST NOT BE ADDED. CONFIRMED PLANET COUNTS HERE AND IN THE TARGET LIST COME FROM THE NASA EXOPLANET ARCHIVE **PSCOMP** TABLE (AKESON ET AL. 2013), ACCESSED 2026 OCTOBER 4. PART TWO, TABLE A2, CARRIES THE EVENTS AND THE CONFIRMATION EXTENTS.

Set of execution blocks	Blocks	Windows	Stars	Systems	Hours	Results it enters
1 Archival scope: blocks the query exposes	656	—	—	—	—	rows 1.1 and 1.2
1.1 ... processed with the frozen pipeline	479	—	—	—	—	panel 2
1.2 ... not processed at the freeze	177	—	—	—	—	panel 3
2 Processed with the frozen pipeline ^c	484	—	—	—	—	rows 2.1 to 2.4
2.1 Primary census	404	1651	89	82	1094	every limit, the crossing list, every chance expectation
2.1.1 Class A, fine channels ^a	—	402	65	60	227	the drifting-carrier experiment
2.1.2 Class B, coarse channels ^a	—	1249	88	81	867	the spectral-excess experiment
2.2 Reserved hold-out	77	315	36	—	214	the rank displacement, the false-alarm tail factor, the radial control profile
2.3 Withheld: primary-beam form invalid ^d	2	8	—	—	—	nothing; withheld
2.4 ϵ Eri at the pointing centre ^e	1	4	1	1	0.1	reported limits only; no census statistic
3 Not processed at the freeze: what became of each	177	—	—	—	—	rows 3.1 to 3.5, the five fates
3.1 Calibrated and searched	144	—	—	—	—	rows 3.1.1 to 3.1.4
3.1.1 Archival extension , reported	115	463	16	16	269	recurrence, and the null on data the frozen pipeline had not seen
3.1.2 Searched after the extension was frozen	8	32	—	—	—	nothing; later than the reported extension
3.1.3 Withheld: primary-beam form invalid ^d	8	32	—	—	—	nothing; withheld
3.1.4 No window reached the export	13	0	—	—	—	nothing
3.2 No calibration in the ALMA delivery	18	—	—	—	—	nothing; the archival gap
3.3 Calibration failed	11	—	—	—	—	nothing; the archival gap
3.4 Not attempted	3	—	—	—	—	nothing; the archival gap
3.5 Working disk insufficient	1	—	—	—	—	nothing; the archival gap
4 Beyond 40 pc: out-of-sample control	47	198	22	—	88	the external null; no science count
5 External calibration sample ^b	326	1322	56	—	—	the rank calibration and the positive control; no science count

^aThe two classes partition the census windows, not its blocks: 252 blocks carry a Class A window and 389 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.

^bNot an out-of-sample set and not an addition to this ledger: every one of its 326 blocks is counted in a row above, 263 in the primary census and 63 in the hold-out, and it is listed apart only to keep it from being counted twice. One further block of the harvest it came from lies outside the 40 pc work list and is excluded here.

^cPanels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 3 of the 5 are in the primary census.

^dEvery one is an ϵ Eri Band 6 block, and the band was withheld whole: in most of them the star sits far enough off axis (response 0.28–0.35, below the 0.5 floor) that the Gaussian primary-beam form is not valid.

^eThe windows of the same band that place the star at the pointing centre, at a response of 1.00, and so pass the floor; their limits are in Appendix A and they enter no census statistic. Row 3.1.1 holds a further 16 such windows, which enter nothing either.

the two noises — has median 1.001 (star-clustered 95 per cent 0.994–1.004), so a stellar-position noise denominator would move no statistic here either.

B.1. The reserved hold-out, and what it calibrates

The statistic and the mask were settled on the searched sample and cannot also calibrate themselves, so a hold-out was reserved by a rule fixed and recorded before any reserved block was searched: a block is held out if and only if $\text{sha256}(\text{UID})_{[8]} \bmod 5 = 0$, subject to two guards that use identifiers alone — a block on the pre-registered list stays in the survey, and no system loses every block — so that the assignment follows from the UIDs and that list alone. The statistic, the trigger and the recurrence criterion were all fixed before the rule itself, and the first reserved block was searched 3 days later, so every diagnostic built afterwards is incapable of promoting a window. Of 484 blocks processed, 77 are reserved (15.9 per cent), giving 315 windows toward 36 stars with no star and no system lost from the headline sample. Three quantities are calibrated on those blocks and on nothing

else.

First, the rank displacement. Out of sample the stellar rank has median 0.405 where one half is expected, with a block-clustered 95 per cent interval of 0.341–0.458 over 315 windows, and 14 per cent of ranks below 0.1 against 6 per cent above 0.9, so the distribution is displaced rather than growing a tail. The displacement runs the way that matters: a rank below one half places the star high, and the star in fact outranks all 512 probes in 14 of the survey’s windows against 3.2 if the positions were interchangeable. The direction therefore inflates star-first outcomes rather than suppressing them, which is why the chance expectations of Appendix D are calibrated on the measured rate. Clustering removes neither the shift nor its size: 99 of 149 blocks of the epoch-extension campaign have a mean rank below 0.5, and a star-clustered bootstrap places the median at 0.440 (95 per cent 0.42–0.46). The searched sample leans the same way but by less (median 0.49, star-clustered 95 per cent 0.47–0.51), and the two intervals do not overlap, so the displacement is a property of these reserved blocks and is not repro-

TABLE A2

THE SURVEY LEDGER, PART TWO: EVENTS, DISPOSITIONS AND CONFIRMATION. EVERY HEADLINE COUNT IN THIS PAPER IS READ FROM HERE. THREE COUNTS ARE 52 OVER THREE DIFFERENT SETS — THE CLASS A CROSSINGS OF THE ADOPTED LEDGER, THOSE OF THE EXTRACTION CATALOGUE, AND THE QUALITY-PASSING CROSSINGS. THE FIRST TWO HOLD DIFFERENT ROWS, SINCE RE-EXTRACTION MOVED FOUR CROSSINGS ACROSS THE TRIGGER EACH WAY; THE LAST TWO HOLD THE SAME ROWS, BECAUSE THE ONE BLOCK FAILING THE QUALITY CRITERION IS ALSO THE SURVEY’S ONLY BLOCK WITH A COARSE-CHANNEL CROSSING. ROW 4.2 IS WHAT THIS PAPER MEANS BY CONFIRMATION AND THE ONLY SUCH COUNT IT QUOTES: A REPEAT EPOCH AT A FREQUENCY THE FIRST DID NOT COVER CONFIRMS NOTHING. ROW 2.2.1 WAS ALSO DECOMPOSED SYSTEM BY SYSTEM, EACH EXPECTATION FORMED FROM THAT SYSTEM’S OWN CONTROL POSITIONS: 2 OF THE 60 SYSTEMS FALL BELOW ONE CHANCE IN TWENTY, AGAINST 3 EXPECTED, AND HD 14055, WHICH HOLDS 8 OF THE 56 CROSSINGS, 8 OF THEM UNATTRIBUTED, GIVES 6 IN CENSUS WINDOWS AGAINST 4.7 EXPECTED OVER ITS 38 CLASS A WINDOWS. NO SYSTEM DEPARTS FROM ITS OWN CONTROLS BY MORE THAN CHANCE ALLOWS.

Count	Value	What it counts
1 Threshold crossings at $T_\star \geq 5$	56	census and archival tail
1.1 Class A, fine channels	52	the carrier experiment
1.1.1 ... in the primary census	50	catalogued window by window
1.1.2 ... in archival-tail blocks	2	both toward HD 14055; in no expectation
1.2 Class B, coarse channels	4	all in the one block failing §5.3.2
2 Quality-passing crossings	52	the same rows as 1.1
2.1 ... attributed to a transition	18	inside the mask in the star’s own frame
2.2 ... unattributed	34	rows 2.2.1 and 2.2.2
2.2.1 ... in the primary census	32	against 28.9 expected by chance, measured at the 402 windows’ own control positions (12–48 allowed)
2.2.2 ... in archival-tail blocks	2	outside that comparison
2.2.3 ... above all 512 controls	2	the two events of Fig. 7
2.2.4 ... recovered in a later block	0	no confirmed signal; 13 attributed ones do
3 Class A systems	60	of 82 surveyed
3.1 ... seen again after 1 day	39	at any frequency
3.2 ... at the same stellar-frame frequency	37	where a signal could have been confirmed ; 11 of them over a year later
3.3 ... with no such second block	23	confirmation impossible; 36 of all 82
4 Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1 ... removed by the line mask	6.4 %	3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union

duced in sample. Either way it runs in the direction that inflates star-first outcomes, which is the quantitative reason the rank is read as a diagnostic and never as a significance.

Second, the false-alarm tail factor — the measured rate at which a no-signal position ranks first, divided by the rate exchangeability would give. It is 1.2 on 5 040 trials, with a bootstrap clustered over the 77 reserved blocks giving 0.6–1.9. It is a statement about the rank and not about the trigger, so it is reported as a diagnostic; the quantity the chance expectations need is the exceedance rate at the stellar position, measured on the same blocks in Appendix D.

Third, the radial behaviour of the control ensemble: the standardised level runs +0.12, +0.10, +0.12, +0.16

TABLE A3

THE 65 ALMA PROPOSALS THE SEARCHED EXECUTION BLOCKS BELONG TO, IN THE STANDARD ADS/JAO.ALMA#code FORM. EVERY BLOCK OF THE PRIMARY CENSUS BELONGS TO ONE OF THESE, AND THE BLOCKS THEMSELVES ARE RETRIEVED FROM THE ALMA SCIENCE ARCHIVE BY THE EXECUTION-BLOCK IDENTIFIERS TABLES D1 AND D2 GIVE.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

across the annulus, a mean per-window slope of $+0.067 \pm 0.033$, small and of the wrong sign to account for the displacement.

The *external calibration sample* must not be confused with the hold-out, and it is not an out-of-sample set. Its 326 execution blocks were reserved to calibrate the rank and to supply repeats the hold-out does not have, and 263 of them are primary-census blocks and 63 hold-out blocks, with 0 outside both: it enters no count of the science sample because its blocks are already counted there, not because it is external. Over its 1322 windows the frozen pipeline returns 7 crossings outranking all their controls — 4 of them β Pictoris CO — and 3 unattributed against 3.7 expected at the measured tail rate.

B.2. What the recurrence statistic reads when nothing is there

The exclusion in Table D1 is one formula. Writing S and σ_d for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\text{excl}} = \frac{S - \bar{a}}{\sqrt{\sigma_d^2 + \bar{\sigma}^2}} = \frac{T_{\text{pers}} - T_{\text{rep}}}{\sqrt{1 + (T_{\text{pers}}/T_\star)^2}}, \quad (\text{B1})$$

with $T_{\text{pers}} = S/\bar{\sigma}$ and $T_{\text{rep}} = \bar{a}/\bar{\sigma}$, so every exclusion follows from three columns of that table and nothing else. T_{rep} is read at the carried drift and at one cell; the largest value over every drift trial at the same cell is tabulated beside it, and because it is a maximum over of order 10^2 trials it is positive on average whatever is there and is never the quantity T_{pers} is compared with. Because the discovery amplitude carries its own error, no exclusion can exceed the strength of the discovery it is built from — and that is also why it is not the strongest statement available. An amplitude selected as the largest of 10^2 to 10^6 cells at $T_\star \simeq 5$ is biased high, so a real carrier would be fainter than measured, and the quantity that survives the bias is the limit $(T_{\text{rep}} + 5)/T_{\text{pers}}$: the fraction of the discovery amplitude still ruled out at the trigger.

The same three numbers say where the test has power and where it has none. The criterion asks the repeat

to reach the trigger, and the reading there is centred on zero with unit scatter, so a carrier radiating through both observations at the discovery amplitude satisfies it with probability $\Phi(T_{\text{pers}} - 5)$, and at the trigger amplitude $5S/T_*$ — the faintest carrier that could have produced the crossing at all — with $\Phi(5T_{\text{pers}}/T_* - 5)$. Both follow from columns Table D1 already prints, and both are tabulated for the two events of Table 6. Where they are small the crossing is untested rather than excluded, and it is reported that way.

The statistic read at the predicted channel is centred on zero and scatters by one, measured and not merely designed. Over 511 single-cell readings of every repeat block in the census and the hold-out its mean is $+0.014$ and its scatter 1.046; at the neighbouring channels of the 2 crossings whose repeat spectra survive — random cells, same registration, same drift — $+0.071$ and 0.966 over 638 cells, with 3.19 the largest value anywhere within $\pm 15 \text{ km s}^{-1}$ of their predicted channels; and at the control positions of the discovery windows -0.036 and 1.034 . The largest of the 8 retained control probes of each repeat block averages 1.449 against the 1.424 expected for a maximum of 8, which is the point: **a maximum is positive on average whatever is there**. The largest of the n_{rep} per-block readings averages $+1.102$ over the crossings carrying exclusions, where the null above predicts $+1.130$ for the same n_{rep} , while the combined T_{rep} of Eq. B1 averages -0.119 . Widening the window costs trials and not calibration: the exceedances of $\$5.6$ match what the same accounting predicts, and HD 69830’s windows are the only ones too narrow for the excursion their crossing’s drift implies.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask exists to say which crossings those are. It reports a frequency coincidence first and an astrophysical judgement about the coinciding species separately, so that the reproducible part can be recomputed and the arguable part argued with.

The list, and when it was made. The mask holds every transition of the 12 species these spectral setups cover — the three CO isotopologues, HCN, HCO^+ , CS, CN, SiO, H_2CO , SO, [C I] and hydrogen recombination — that falls inside one of the 46 searched frequency intervals with an upper-state energy below 150 K . That rule is the whole of it, and it admits 111 transitions (Table C1). No strength cut is applied across species, since any threshold admitting the strong lines of HCN would discard the brightest lines in these fields; within a species the main hyperfine components are kept and the weak satellites dropped. The list is a single Splatalogue query of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) made on 2026 October 6, after the disposition criteria were fixed on 2026 September 11, and frozen before any crossing was compared with it.

The frame is the star’s own, and the sign is not the usual one. A crossing’s frequency is carried from the topocentric value to the barycentre using its own block’s pointing and mid-time, and then to the stellar rest frame using the star’s systemic velocity. The barycentric term alone spans 58 km s^{-1} across the survey, so a tolerance

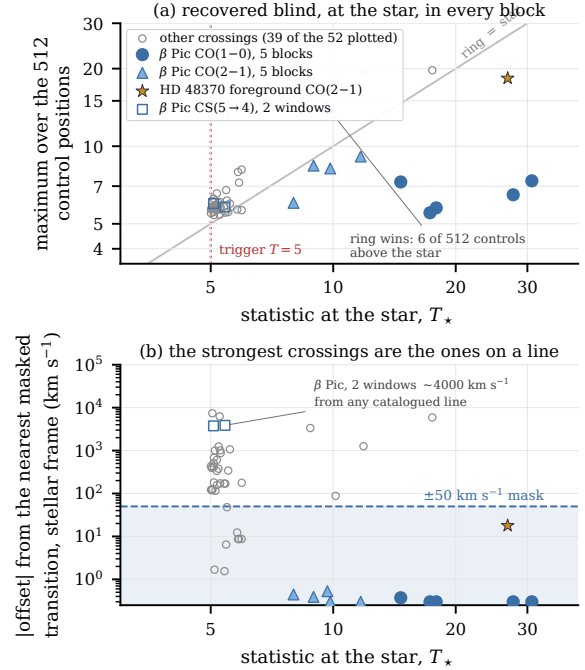


FIG. B1.— The β Pictoris positive control, in both directions. Both panels draw the same 52 of the 56 crossings, the remainder being crossings for which no control ring was retained. (a) The spatial screen: the stellar statistic against the maximum over that window’s 512 control positions. Filled symbols are the blocks in which the blind search recovers circumstellar carbon monoxide at β Pictoris, 9 of them with the star above every control and one with the ring above it, that one being block X7116f1_X1785, Band 6, where the ring reaches 14.13 against 9.68 at the star (§5.4). (b) The same crossings by velocity offset from the nearest masked transition in their own star’s rest frame, with the adopted $\pm 50 \text{ km s}^{-1}$ half-width marked. The 10 β Pictoris carbon-monoxide crossings land $0.1\text{--}0.5 \text{ km s}^{-1}$ from CO; its 2 CS($5\rightarrow 4$) crossings sit $3765\text{--}3870 \text{ km s}^{-1}$ from anything catalogued. 39 further crossings are drawn in grey, with HD 48370 marked.

applied in the observed frame would spend most of its budget on the Earth’s motion. The 3 HD 285968 epochs show the transform working in the only way that can be checked: 12 km s^{-1} apart as observed, they agree to 0.03 km s^{-1} in the star’s frame. Toward HD 139084 B the barycentric and systemic terms nearly cancel, $+4.5161$ against $+4.5500 \text{ km s}^{-1}$, so that row’s two frequencies agree to 39 kHz although both transforms were applied. Throughout, $\Delta v_* = c(\nu_* - \nu_0)/\nu_0$, so a positive offset means the crossing lies above the transition in frequency — the opposite sign to the velocity convention of a spectral line.

Interstellar gas is at rest in a different frame, so the mask is evaluated in two. Circumstellar gas shares its star’s motion, but gas along the line of sight is at rest in the local standard of rest, so the barycentric frequency is carried to that frame as well — one further step, the projection of the standard solar motion — and a crossing is attributed if it falls inside the mask in either frame. The two offsets differ by the star’s own LSR velocity and by nothing else, which here runs $0.3\text{--}18.7 \text{ km s}^{-1}$ about a median of 3.0 , the two published conventions of the LSR differing by 0.29 km s^{-1} . In this sample no crossing changes side, and none could have: the unattributed crossing nearest a transition lies 88 km s^{-1} from one, 38 km s^{-1} beyond the mask and twice the largest of those velocities. In the other direction one coincidence,

TABLE C1

THE LINE MASK, BY SPECIES: THE NUMBER OF TRANSITIONS N IT HOLDS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES. HYDROGEN RECOMBINATION LINES CARRY NO TABULATED E_u . THE CRITERION AND THE DATED QUERY ABOVE DEFINE THE LIST EXACTLY, SO THE INDIVIDUAL TRANSITIONS CAN BE RECOVERED FROM THE LINE CATALOGUES THEMSELVES.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03–231.90	–
SO	15	100.03–682.68	16–143
H ₂ CO	13	101.33–491.97	10–106
C ¹⁸ O	3	109.78–329.33	5–32
CN	50	113.12–680.30	5–114
CO	3	115.27–345.80	6–33
CS	3	195.95–342.88	24–66
SiO	2	217.10–347.33	31–75
¹³ CO	8	220.40–330.59	16–32
HCN	7	354.50–354.51	43–43
HCO ⁺	1	356.73–356.73	43
[C I]	1	492.16–492.16	24

HD 161868’s -48.0 km s^{-1} from SO(2_3-2_1), falls outside the mask in the LSR frame at -53.4 , which is why the frames are taken together rather than exchanged.

What the second frame changes is which attributions are interstellar. Of the 24 stars carrying crossings, the composite CO($1 \rightarrow 0$) survey of Dame & Thaddeus (2022), which contains Dame et al. (2001), covers 11 and holds emission toward 5 of them, between -18 and $+22 \text{ km s}^{-1}$; the all-sky Planck CO map (Planck Collaboration et al. 2016) stays below $0.178 \text{ K km s}^{-1}$ toward every other, including the 13 outside the CfA footprint. Toward ALMA J153702653–33192492, HD 139084 B, HD 285968 and HD 48370 the CO spectrum is non-zero at the crossing’s own LSR velocity, within 3 km s^{-1} of its own peak and at 0.23 K or brighter against a 0.18 K channel: 6 of the 21 coincidences are interstellar rather than circumstellar. The clearest is ALMA J153702653–33192492, toward the Lupus clouds: its CO($2 \rightarrow 1$) crossing lies 0.5 km s^{-1} from a 6.6 K km s^{-1} cloud line at $+4.9 \text{ km s}^{-1}$, and the star is at 35.0 pc with the clouds far behind it, so the field is a molecular sight-line and not a disc. It is the velocity coincidence that says so and not the field: the brightest control position does stand above the star in all six of its windows, 13.4 against 5.4 in the CO($2 \rightarrow 1$) one, but so it does in 390 of the 402 Class A windows of the survey, and the comparison therefore distinguishes nothing. Its other 5 crossings lie at least 128 km s^{-1} from every transition of the list in both frames, and the mask disposes of none of them.

The coincidences, and the one species we do not read as emission. Of the 56 crossings, 21 fall within $\pm 50 \text{ km s}^{-1}$ of a transition of the list in their own star’s frame; that count is the first layer and all the velocity criterion decides. Coincidences with sulphur monoxide account for 3 — CD–57 1054, CP–72 2713 and HD 161868, at -1.7 , -12.3 and -48.0 km s^{-1} from transitions of upper-state energy $21-87 \text{ K}$, each inside a Band 7 window that also covers CO($3 \rightarrow 2$) — and two things argue against reading those as emission. Sulphur monoxide has not been reported in a debris disc, where the second-generation gas is CO, C and O; and toward CP–72 2713 the CO($3 \rightarrow 2$) line falls in the same window and is absent to 10.5 mJy in a 0.42 km s^{-1} channel while the crossing itself is 12.2 mJy , so an SO line there would have to outshine an undetected CO($3 \rightarrow 2$) line in the same data. Both arguments are

about the species and neither about any one crossing, so all 3 are treated alike: the offset is printed beside each in Table D1, where a dagger marks the rows this applies to, and disposes of none of them, giving 18 attributed against 38 unattributed, of which two outrank every control position in their own windows. Reading them as attributions instead gives 21 and 35, and 1 such crossing; the recurrence test excludes a persisting carrier for all 3 either way, at $4.1-5.5\sigma$, so the choice moves counts and no conclusion. The reserved hold-out holds 2 coincidences, 1 of them the same species and treated the same way, leaving 1 attributed and 7 unattributed there (§5.6.2).

The half-width, and what it costs. A one-channel test is too tight for circumstellar gas, whose emission is shifted by many channels: the width follows from Keplerian motion where these molecules survive, $\lesssim 13 \text{ km s}^{-1}$ beyond 10 au , and is $\pm 50 \text{ km s}^{-1}$. At that width the mask removes 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent of it, against 5.04 GHz of the 118.1 GHz both classes cover, 4.3 per cent: the cost falls on the primary experiment, because its windows are the line-tuned ones. Evaluating the mask in two frames widens the masked band to the union of two tubes separated by the star’s LSR velocity and raises the Class A cost to 3.72 GHz , 7.8 per cent — 22 per cent more bandwidth, which buys the 6 interstellar identifications above and no further attribution. Between ± 20 and $\pm 100 \text{ km s}^{-1}$ the mask labels 17 to 19 of the crossings, and the 2 that outrank all 512 control positions of their own windows are the same two at every width.

What a bright line costs is its channels, and nothing beyond them. Because every above-trigger cell of a window is recorded (§5.3), a weaker carrier beside a molecular line is recorded too, and the price of line confusion is the masked channels and not the windows holding them. Only the 3 fine windows narrower than the mask itself are lost outright, 0.108 GHz , 0.23 per cent of the Class A union. The one masked window that still holds a trigger-level cell outside the mask is β Pic at 115.6200 GHz , $+927 \text{ km s}^{-1}$ from the nearest transition, and it does not recur: the 4 other blocks covering that cell read $T_\star = 1.7$ to 3.4 .

The labelled crossings are tested for recurrence too. Of the 18 attributed crossings, 15 have a later block covering the predicted cell and 13 are present again, toward β Pic and HD 285968 at $T_\star = 5.9$ to 30.8 — the test working in the positive direction on emission that is certainly real, and no exclusion is formed for those rows, because what it would exclude has just been observed. For the 2 not recovered a repeat window’s own largest statistic bounds the predicted cell from above, and Table D1 prints the row as a bound; 3 have no covering block at all (HD 139084 B, HD 48370 and HIP 10679).

Image-sideband leakage accounts for no crossing. The mask holds signal-sideband frequencies, and the 2 unattributed crossings that share a block with a CO($2 \rightarrow 1$) crossing in the opposite sideband would need a first local oscillator outside the range their own block’s spectral windows permit, $235.02-235.68 \text{ GHz}$; their image frequencies fall at least 1353 km s^{-1} from every masked transition.

A longer list is not a better one. Every Splatalogue transition over these intervals with an Einstein coeffi-

cient above 10^{-5} s^{-1} — 5,114 transitions of 235 species — would mask 75.1 per cent of the union and move 27 crossings across the line, one of them rank-leading. At these frequencies almost anything lies within $\pm 50 \text{ km s}^{-1}$ of something in such a list, which is why the species are restricted to what these setups cover and why the outcome is a label rather than a verdict.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

This appendix gives the chance expectation for the unattributed crossings, what it is conditioned on, and what the control ensemble calibrating it can support.

D.1. *What the expectation is conditioned on*

The expectation is built from the control positions, the only empirical null the data provide, and each Class A window contributes two quantities measured on itself. The fraction of its own 512 control positions that reaches the trigger is the probability that a position in that field does so — no cell count enters it and no form is assumed for the tail — and summed over the windows 36.1 are expected to trigger by chance. The share of its own channel grid lying inside an attribution window, in that star’s rest frame, is the probability that such a position would be attributed, a control maximum being free to fall on any channel. Weighted by the first that share is 20 per cent, against a median of 6.0 per cent per window, because 15 narrow line-tuned windows lie wholly inside the mask and their fields, which hold real emission, are where a control position most often triggers. What is left is 28.9 expected unattributed Class A crossings against 32 observed.

Both sides run over the 402 Class A windows of the primary census, each contributing its own measured noise and its own masked grid share, so the expectation is a sum over rows and nothing is carried at an assumed rate. A further 15 Class A windows arrived with the archival tail and lie outside it, together with the two unattributed crossings toward HD 14055 they hold, which are reported beside the comparison and enter neither side of it. A systemic velocity is measured for the stars of 277 of the census windows and the others are taken at rest, which moves the masked share by 3.0 points. No window holds more than one crossing, so both sides count windows.

The one assumption the searched data cannot test is that a star behaves like a control position in its own field, and the reserved blocks measure it in the quantity used here. Their stars reach the trigger in 8 of 62 fine-channel windows where the control positions of those same windows predict 7.5: a ratio of 1.07, with 0.48–1.79 allowed by resampling the reserved blocks, and no two of the events in one block. In the 253 coarse-channel reserved windows no star reaches the trigger, against 0.04 predicted. The same blocks place the stellar rank below one half (Appendix B), which is the same departure read in the body of the distribution rather than in its tail and runs the same way. Carrying the measured ratio through to the expectation gives 30.8 against the 28.9 above, and the observed 32 sits inside the range either way; the ratio is itself consistent with unity, so the uncorrected sum is the one quoted and this is the size of the correction it leaves out.

The 12–48 quoted beside the expectation is what 20 000

resamplings of the execution blocks allow, each window’s rate redrawn from its own 512 probes: windows share blocks and are not independent trials, and the ensemble is finite. Those two terms alone give 26–32, so what widens the range is the calibration above. Read as the interval it is, it leaves the expectation anywhere between 12 and 48: an added population of as many as 19 events would be absorbed by that width, and that, rather than any smaller figure, is the limit on what the comparison can exclude.

Every Class B crossing in the survey comes from the one execution block of §D.2, and setting it aside the coarse class adds almost nothing to either side: over both classes together 29.0 are expected against the same 32 observed. Only 455 of the 1651 windows hold any control position above $T_{\text{trig}} = 5$ at all, a coarse single pointing having too few cells for its own field to reach the trigger by chance, so the Class A comparison is also the survey’s.

The statistic is always the one measured over the window’s whole field. 157 windows admit integrations lying inside the primary beam that a narrower field would discard, which recovers a median 2.1 times the on-source time and moves the search statistic by a median 0.07; the whole-field value is the one adopted, on 8 rows of Table D1, and a window’s control ensemble is measured with it, so neither side of the comparison mixes two fields. Where the extraction catalogue’s disposition column disagrees — on 4 crossings that fall below the trigger, and the species list moves none into attribution — the ledger is the authority.

D.2. *Execution-block quality*

The comparison with the control positions assumes each block’s ensemble behaves like noise, and one block makes that false: every one of its 512 probes exceeds the trigger, the smallest at 6.6. A criterion for the condition is therefore applied to every block rather than to the one that prompted it.

Two criteria are built from that statistic and they are not the same test. The statistic is the median of a window’s own 512-position control ensemble, which separates a field with a source in it from a field that is no null at all. A *window* is flagged when that median exceeds twice the median of the same quantity over the survey, 5.5 against 2.74; the flag is a property of the window and cuts across attribution by design, firing on 5 of the 1651 windows in 2 blocks toward 2 stars (BD+05 1668 and HD 48370), and every one of those windows carries a crossing, attributed and unattributed alike. A *block* is judged on the same median divided by the median over the windows of its own class, since a maximum over a larger channel×drift grid sits higher and one threshold should mean the same thing in both; a block fails when the median of that ratio over its windows exceeds twice the scale of its class. The factor is the same in both. Neither was fixed in advance: the condition was noticed in one block and the criterion written afterwards, so it is post hoc, and what makes it trustworthy is that it does not depend on where the line was drawn — the failing block sits at 4.1 and the next worst at 1.12, so any threshold between them selects the same single block.

Applied to all 404 blocks of the primary census, where the median block sits at 1.00, the criterion fails 1: A002_Xc7fa6f_X28a8 toward BD+05 1668. Nothing

else is close, and the unscaled form of the rule fails the same 1 block. Its 4 windows are all coarse-channel and hold the survey’s entire Class B crossing population, so it carries 4 of the 56 threshold crossings — 0 of them outranking all 512 control positions of their own window — and no Class A statistic here has ever contained it. Those 4 appear in Table D1 with their block named and enter no count elsewhere.

D.3. *The control ensemble: its calibration, and its resolution*

The ring geometry is set in §4.1; its calibration and its resolution are the subject here. Over the 1645 census windows that retain a control vector the stellar ranks are consistent with the one half exchangeability implies — median 0.48 with each star carrying equal weight, 0.47 to 0.51 — in both array strata, the compact-array windows measured on each observation’s own antenna table rather than on a 12 m beam. That is a weak statement and not a strong one, since a shift smaller than about 0.042 could not have been seen in a sample of this size. The reserved blocks lean further from one half, and Appendix B.1 compares the two samples. The expectations above are summed from the exceedance rate measured at each window’s control positions, so they do rest on that transfer and carry its measured accuracy.

The rank is not a significance test, and the reason is structural. Ranking the star against 512 control positions of the same field decides which windows deserve examination, but $1/(512+1) = 1.95 \times 10^{-3}$ is the *resolution* of the ensemble — the smallest rank it can express — and not the probability that noise produced the window. A survey of 1651 windows would need a rank 64 times finer, and correlation makes it coarser still: controls inside the synthesised beam are not independent, so an annulus resolves no better than 0.023–0.032 however many probes are drawn into it, with the primary beam bounding how many a single pointing contains. That floor is the design’s central limitation: no single window here can be significant on the rank alone, which is why confirmation requires independent recurrence.

D.4. *The trials budget*

The searched cells counted in §4.2 are not independent trials: channel smoothing and a drift grid stepped twice per channel correlate neighbouring ones, which alone would leave 0.19 of them independent. The control ensembles say otherwise, their maxima sitting at 0.995 of the prediction for the maximum of the full searched count treated as independent and Gaussian — a budget predicting 476 windows with a control above T_{trig} against the 455 observed — while fitting each window’s effective count against its own ensemble returns 0.81 of its searched cells. Nothing turns on which is right, because the expectation above is summed from measured rates and not from a cell count: by the cell-count route the number reaching the trigger is 36.5 rather than 36.1, 3 per cent away.

The on-star crossings do exceed that budget, and the attribution accounts for it. A Gaussian tail over every searched cell predicts 37.8 stellar exceedances against the 56 found, but 18 are identified with a catalogued transition, 4 come from the block of §D.2 and 2 fall outside the

census windows, leaving 32 against the 29.0 predicted. No excess survives the attribution.

One stratum departs from uniformity, and the departure is where the line emission is. Counting the windows whose stellar peak both reaches the trigger and exceeds all 512 control positions of its own window, the 402 fine-channel windows hold 12 against 0.78 expected, while the coarse stratum, three times larger and predicting 2.43, contains 0. A coarse channel dilutes a narrow circumstellar line, so the class that can resolve one is the class that finds it — and an unresolved carrier would appear in Class A first for the same reason. Reaching the trigger is holding a crossing, so each of those windows holds one and 10 hold a crossing the mask attributes: the departure is a statement about the windows carrying line emission and not about the fine stratum at large. Drop the trigger condition and the star leads its own ring in 2 further windows, at 4.20 and 4.36, giving 14 over all 1651 windows against 3.2 expected. One quantity, two conditions; the chance comparison uses the conditioned one.

What is left is bounded, and the bounds are null. The real stellar ranks agree at $p = 0.38$ with the 220 672 ranks each control takes against the other probes of its own window; radial suppression is below 10 per cent in the mean; and scrambling a window so nothing coherent survives along a drift track carries 4 past the rank floor and changes no disposition.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing, so the screening this survey has, and the coverage it does not have, are both set out here. Millimetre interferometry is an unusually poor environment for terrestrial interference, though not an empty one: ITU Radio Regulations Article 5 (International Telecommunication Union 2024) carries active allocations throughout 90–275 GHz and WRC-19 identified parts of 275–450 GHz for land-mobile and fixed use. The specific hazard is the 94.0–94.1 GHz Earth exploration-satellite allocation used by spaceborne cloud radar (Stephens et al. 2002; Wehr et al. 2023), a documented contaminant of ALMA Band 3; 1 of the 1651 census windows overlaps it, carrying 0 threshold crossings and 0 rank-passing events, so no result here rests on it. What favours this band is that terrestrial transmitters are sparse compared with centimetre wavelengths and, decisively, that an interferometer is not a total-power detector: a signal not in the far field at the phase centre does not fringe-track but acquires a rapidly varying geometric phase across the array, and is suppressed in the correlation rather than imaged.

The coverage of the screening is uneven, and that is stated rather than smoothed over. The allocation test above and the sky- and intermediate-frequency tests below are computed for every searched window of the census, as is the 512-position rank screen, which asks whether an excess is at the star or everywhere in the field. Recurrence reaches only the crossings with a covering repeat block, and the visibility fit is a consistency check rather than a filter. The correlator flagging is the observatory’s own, replayed from each block’s delivered

TABLE D1

THE CROSSING LEDGER, FROM WHICH EVERY DISPOSITION IN THIS PAPER CAN BE REPRODUCED. ALL 56 THRESHOLD CROSSINGS, INCLUDING THE 4 IN THE FAILING BLOCK OF §5.3.2. EVERY ROW IS THE PRIMARY CROSSING OF ITS WINDOW; THE 21 SECONDARY CROSSINGS ARE NOT LISTED INDIVIDUALLY, AND THE THREE OF THEM THE MASK DOES NOT ATTRIBUTE ARE DESCRIBED IN §5.3. THE COLUMNS ARE THE STAR, ITS EXECUTION BLOCK, THE ALMA BAND, THE CROSSING FREQUENCY AS THE CORRELATOR DELIVERED IT AND THE SAME FREQUENCY IN THE STAR'S REST FRAME, THE SEARCH STATISTIC T_* , THE LARGEST STATISTIC T_{ctrl} AMONG THE WINDOW'S 512 CONTROL POSITIONS, THE NEAREST MASKED TRANSITION AND THE STELLAR-FRAME OFFSET Δv_* FROM IT, THEN THE RECURRENCE TEST: n_{rep} COVERING REPEAT BLOCKS SPANNING Δt DAYS, THE STATISTIC T_{pers} A CARRIER PERSISTING AT THE DISCOVERY FLUX WOULD HAVE PRODUCED AT THE PREDICTED STELLAR-FRAME CELL, THE STATISTIC T_{rep} MEASURED THERE AT THE CARRIED DRIFT, AND THE EXCLUSION OF EQ. B1, WHICH THREE OF THESE COLUMNS REPRODUCE, TWO OF THEM ALSO GIVING THE LIMIT DEFINED IN APPENDIX B. WHERE THE EMISSION IS PRESENT AGAIN IN A COVERING BLOCK THE TEST RETURNS A RECOVERY AND NOT AN EXCLUSION, SO BOTH T_{rep} AND σ_{excl} ARE BLANK AND THE DISPOSITION SAYS WHICH IT WAS. T_* AND T_{ctrl} COME FROM THE SAME MEASUREMENT OF THE SAME WINDOW; T_{ctrl} GATES NOTHING AND SAYS ONLY WHICH CROSSING WAS EXAMINED FIRST, AND THE DRIFT-MAXIMISED READING OF THE CELL IS GIVEN IN TABLE 6 FOR THE TWO LEADING EVENTS AND IS NOT TABULATED FOR THE REST. ROWS WHOSE WINDOW IS DATA-QUALITY FLAGGED (§D.2) ARE MARKED, AND ON THOSE T_{pers} AND THE EXCLUSION ARE WITHHELD: BOTH ARE BUILT FROM THE DISCOVERY AMPLITUDE, AND IN A WINDOW WHOSE CONTROL RING STANDS AS HIGH AS ITS STAR THAT AMPLITUDE CANNOT BE ASSIGNED TO A POINT SOURCE AT THE STAR. 1 OF THOSE ROWS LIES OUTSIDE THE FAILING BLOCK — HD 48370, A RESOLVED DISC — AND IS WITHHELD FOR THAT REASON AND NOT FOR ITS BLOCK. A CROSSING IS ATTRIBUTED WHERE $|\Delta v_*| \leq 50 \text{ km s}^{-1}$ AND THE COINCIDING SPECIES IS ONE SUCH A SYSTEM COULD PLAUSIBLY EMIT; A DAGGER MARKS THE 3 ROWS THAT SATISFY THE FIRST CONDITION AND NOT THE SECOND, FOR THE REASON GIVEN IN APPENDIX C. THE TRANSITION NAMED IS THE NEAREST IN THAT FRAME, SO FAR FROM EVERY TRANSITION IT IS THE MAGNITUDE AND NOT THE NAME THAT CARRIES INFORMATION. BLOCK IDENTIFIERS OMIT THE CONSTANT A002_ PREFIX.

Star	block	B ν_{topo} (GHz)	ν_* (GHz)	$T_* T_{\text{ctrl}}$ line	$\Delta v_* n_{\text{rep}}$	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
BD+05 1668	Xc7fa6f_X28a8	6	240.1016	240.1054 73.398 74.69 CS(5-4)	-5911.9	29	0.9-113	n/a	+0.15	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	224.7422	224.7458 48.303 83.74 H ₂ CO(3 _{1,2} -2 _{1,1})	-1264.5	29	0.9-113	n/a	+0.37	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	226.7422	226.7458 41.355 75.48 CN(2 _{3/2} -1 _{1/2})	+88.0	29	0.9-113	n/a	+0.66	n/a unattributed; flagged
BD+05 1668	Xc7fa6f_X28a8	6	242.1953	242.1992 39.586 83.29 CS(5-4)	-3349.2	29	0.9-113	n/a	+0.77	n/a unattributed; flagged
β Pic	Xf5d76d_Xeb8	3	115.2603	115.2711 30.765 7.34 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_Xe19	3	115.2603	115.2711 27.684 6.48 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
HD 48370	Xc26103_X155a	6	230.5117	230.5243 26.834 18.36 CO(2-1)	-17.8	—	—	n/a	—	n/a attributed; flagged
β Pic	Xf4d6f_Xc7	3	115.2620	115.2711 17.907 5.76 CO(1-0)	-0.2	4	53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_Xcc5	3	115.2604	115.2711 17.288 5.52 CO(1-0)	-0.2	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xf5d76d_X32f1	3	115.2606	115.2713 14.654 7.27 CO(1-0)	+0.4	4	0.0-53	n/a	—	n/a attributed; recurs
β Pic	Xd9668b_X3a90	6	230.5161	230.5378 11.681 9.12 CO(2-1)	-0.3	5	1.0-1983	n/a	—	n/a attributed; recurs
β Pic	X6f1341_X1484	6	230.5284	230.5379 9.832 8.20 CO(2-1)	-0.1	5	25-1984	n/a	—	n/a attributed; recurs
β Pic	X7116f1_X1785	6	230.5272	230.5384 9.677 14.13 CO(2-1)	+0.5	5	25-1959	n/a	—	n/a attributed; recurs
β Pic	Xd9668b_Xa9df	6	230.5160	230.5377 9.948 8.41 CO(2-1)	-0.4	5	1.0-1984	n/a	—	n/a attributed; recurs
β Pic	Xa7a216_X23e4	6	230.5273	230.5377 7.988 6.03 CO(2-1)	-0.4	5	601-1313	n/a	—	n/a attributed; recurs
61 Vir	Xc079b5_X82f	7	346.3138	346.3232 5.958 5.65 SO(9 ₈ -8 ₇)	-177.6	3	0.1-3.0	9.7	+0.39	4.9 unattributed; no recurrence
HD 285968	Xd98580_X3bf6	6	230.5021	230.5447 5.956 8.12 CO(2-1)	+8.7	3	13-44	n/a	—	n/a attributed; recurs
HD 285968	Xdb7ab7_X5b4e	6	230.5116	230.5447 5.868 7.22 CO(2-1)	+8.7	3	5.0-44	n/a	—	n/a attributed; recurs
HD 285968	Xdb4b9a_X108f	6	230.5100	230.5447 5.835 7.94 CO(2-1)	+8.7	3	5.0-39	n/a	—	n/a attributed; recurs
CP-72 2713	Xff0235_X4a6d	7	344.2697	344.2965 5.809 5.68 SO(8 ₈ -7 ₇)	-12.3 [†]	5	0.1-2.1	6.3	-0.82	4.8 unattributed; no recurrence
HD 139084	B Xcd8029_Xb6b0	7	345.8241	345.8241 5.763 17.70 CO(3-2)	+24.4	—	—	—	—	attributed
HD 14055	Xff3d8_Xd46	7	331.7699	331.7664 5.575 5.99 ¹³ CO(3-2)	+1068.7	30	0.9-650	170.4	-0.02	5.6 unattributed; no recurrence
HD 23484	X10b22d7_X2445	6	230.7904	230.8006 5.520 5.55 CO(2-1)	+341.5	4	0.1-2.1	10.4	+1.25	4.3 unattributed; no recurrence
HD 161868	Xff45a6_X10ca8	7	345.6338	345.6491 5.481 6.16 SO(2 ₃ -2 ₁)	-48.0 [†]	13	0.0-6.1	92.2	+0.24	5.5 unattributed; no recurrence
HIP 10679	Xa95c04_X14cd	6	220.4185	220.4035 5.456 6.76 ¹³ CO(2-1)	+6.5	—	—	—	—	attributed
HD 14055	X11adad7_X25c5e7	7	345.4564	345.4303 5.448 6.82 SO(2 ₃ -2 ₁)	-237.8	30	0.1-653	14.0	+0.03	5.1 unattributed; no recurrence
HD 207129	Xc19d6f_X5706	6	230.4226	230.4070 5.433 5.52 CO(2-1)	-170.3	4	0.8-425	69.7	-0.93	5.5 unattributed; no recurrence
β Pic	Xd9668b_X3a90	6	241.7512	241.7739 5.420 5.81 CS(5-4)	-3869.7	1	1.0	5.7	+0.72	3.4 unattributed; no recurrence

TABLE D2

The crossing ledger, continued. THE REMAINING 28 OF THE 56 THRESHOLD CROSSINGS, IN THE SAME ORDER AND WITH THE SAME COLUMNS AS TABLE D1, WHOSE CAPTION DEFINES THEM. THESE TOO ARE PRIMARY CROSSINGS, ONE TO A WINDOW.

Star	block	B ν_{topo} (GHz)	ν_* (GHz)	$T_* T_{\text{ctrl}}$ line	$\Delta v_* n_{\text{rep}}$	Δt (d)	T_{pers}	T_{rep}	σ_{excl}	disposition
HR 1010	Xc6fa08_X1053	6	230.5133	230.5287 5.410 5.65 CO(2-1)	-12.0	7	2.0-14	8.4	<+4.60	>2.1 attributed; not recovered
ALMA J153702653-33192492	Xff0235_X8392	6	230.5222	230.5392 5.405 13.41 CO(2-1)	+1.6	6	18-141	6.2	<+4.15	>1.4 attributed; not recovered
HD 110411	Xe45e29_Xb7d4	6	219.4538	219.4357 5.393 5.85 C ¹⁸ O(2-1)	-170.2	1	10	6.7	+0.62	3.8 unattributed; no recurrence
HD 31392	X129754b_X6946	6	231.1998	231.2231 5.289 5.81 H(30 α)	-876.3	8	0.1-41	15.7	-0.26	5.1 unattributed; no recurrence
HD 14055	Xff99e1_X24be0	7	331.7078	331.7035 5.277 5.69 ¹³ CO(3-2)	+1011.6	30	0.9-652	252.2	+0.15	5.3 unattributed; no recurrence
AU Mic	Xa42f75_X319	6	230.6864	230.6705 5.274 6.07 CO(2-1)	+172.3	2310-455	6.4	-0.16	4.2 unattributed; no recurrence	
HD 69830	X12209f7_X16edb7	7	322.4325	322.4445 5.259 5.42 C ¹⁸ O(3-2)	-6268.5	5	0.1-29	8.9	-0.48	4.8 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	216.1844	216.2036 5.228 6.02 SiO(5-4)	-1244.6	7	18-126	11.0	+1.03	4.3 unattributed; no recurrence
AU Mic	X899169_X1838	6	230.8252	230.8294 5.221 5.85 CO(2-1)	+379.0	2145-310	10.0	+0.19	4.5 unattributed; no recurrence	
HD 14055	Xff99e1_X2c3bb	7	345.5256	345.5216 5.186 5.66 SO(2 ₃ -2 ₁)	-158.6	30	0.1-651	172.8	+2.00	5.1 unattributed; no recurrence
HD 14055	Xff99e1_X3835	7	345.0099	345.0070 5.177 5.66 SO(2 ₃ -2 ₁)	-604.8	30	0.1-649	187.2	-1.78	5.2 unattributed; no recurrence
HD 14055	Xff99e1_X2b2ee	7	344.7007	344.6965 5.175 5.92 SO(8 ₇ -7 ₆)	+336.0	30	0.1-651	210.7	-0.00	5.2 unattributed; no recurrence
LHS 1140	Xd9d7c7_X6fd2	6	230.6355	230.6268 5.136 5.68 CO(2-1)	+115.5	4	5.0-27	7.3	+0.33	4.0 unattributed; no recurrence
HD 10647	Xff294f_Xa7f0	7	331.6570	331.6943 5.120 5.91 ¹³ CO(3-2)	+1003.3	10	0.1-592	125.8	+0.39	5.1 unattributed; no recurrence
CD-57 1054	Xc525f_X5aa4	7	346.5099	346.5265 5.108 5.71 SO(9 ₈ -8 ₇)	-1.7 [†]	1	39	6.0	-0.27	4.1 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	217.7292	217.7269 5.091 5.41 H ₂ CO(3 _{0,3} -2 _{0,2})	-680.5	7	3.8-242	11.6	+0.07	4.6 unattributed; no recurrence
GJ 849	Xd9668b_X841a	6	230.8769	230.8581 5.090 5.83 CO(2-1)	+416.3	5	0.1-51	9.3	-1.69	5.3 unattributed; no recurrence
ALMA J153702653-33192492	Xfe62c1_X398	6	218.8729	218.8923 5.090 5.56 H ₂ CO(3 _{2,1} -2 _{2,0})	+181.2	7	18-126	10.5	-1.04	5.0 unattributed; no recurrence
HD 14055	X1001fdc_X29ae	7	331.1559	331.1546 5.082 6.26 ¹³ CO(3-2)	+513.8	—	—	—	—	unattributed; no spectrum
β Pic	Xd9668b_Xa9df	6	241.8367	241.8594 5.079 5.99 CS(5-4)	-3765.1	1	1.0	4.8	-0.63	3.9 unattributed; no recurrence
ALMA J153702653-33192492	Xf8f6a9_X10359	6	218.6711	218.6688 5.048 5.77 H ₂ CO(3 _{2,1} -2 _{2,0})	-125.1	7	3.8-242	11.6	+0.00	4.6 unattributed; no recurrence
HD 69830	X1227ed_X178c6	7	321.2005	321.2149 5.046 6.13 C ¹⁸ O(3-2)	-7387.7	5	6.0-36	10.2	-0.46	4.7 unattributed; no recurrence
HR 1010	Xc66c1_Xaa9	6	230.8348	230.8491 5.044 5.72 CO(2-1)	+404.6	7	1.1-14	10.3	-1.53	5.2 unattributed; no recurrence
61 Vir	Xc067f7_X822	7	345.5599	345.5678 5.038 5.79 SO(2 ₃ -2 ₁)	-118.5	3	2.0-3.0	9.3	+0.45	4.2 unattributed; no recurrence
HD 170773	Xff0235_X8c42	7	331.0101	331.0234 5.029 5.75 ¹³ CO(3-2)	+394.8	4	0.9-113	74.8	+1.56	4.9 unattributed; no recurrence
ALMA J153702653-33192492	Xd9d7c7_Xbb64	7	347.2179	347.1890 5.009 5.68 SiO(8-7)	-122.2	1	0.1	5.0	-2.06	5.0 unattributed; no recurrence
HD 14055	X1190de8_Xd526	7	344.4676	344.4515 5.006 5.96 SO(8 ₈ -7 ₇)	+122.7	30	20-611	11.6	-0.60	4.8 unattributed; no recurrence
η Crv	X122b6f_X1041e	7	357.2815	357.2475 5.006 5.50 HCO ⁺ (4-3)	+431.3	1	87	11.0	-0.62	4.8 unattributed; no recurrence

calibration script, and was not retained window by window. *No per-channel interference flag of our own exists, and none can be recovered.* Four measurements follow, and each is a test the population could have failed.

First, clustering in sky frequency. If interference dominated the crossing population, the crossings would pile up at particular sky frequencies. They do cluster — but on the survey’s own molecular lines. The test needs a measured peak frequency in the frame in which an interfering carrier would stand still, so it runs topocentrically over the 51 of the 56 crossings for which one was measured, against the transitions the search was tuned to. For the other 5, all β Pic carbon-monoxide crossings, only the velocity offset from the line survives, and a frequency reconstructed from a line offset cannot be used to ask whether frequencies crowd onto lines. Of the 51, 11 fall within $\pm 50 \text{ km s}^{-1}$ of a searched transition against 4.7 expected if each window’s peak fell anywhere on that window’s own channel grid, $p = 0.0007$. The densest band, 230.422–230.723 GHz, holds 12 of them, 9 within that velocity window of CO(2→1); and once the line-attributed crossings are set aside, the 40 that remain show no clustering at all ($p = 0.111$). *The concentration is the attribution, not a residue left over after it.* Table D1 counts the same band as 14 crossings with 11 attributed, a different test over a different set — all 56, each attributed in its own star’s rest frame — and the 2 it adds are the ones whose peak frequency this test has no measurement of.

Second, the same test against the atmosphere. A residual of the atmospheric calibration stands still in topocentric frequency exactly as a transmitter does, and it is narrow where the atmosphere has lines, so the crossings can be asked whether they avoid the sky or the air. 570 transitions of ozone, water and oxygen lie within 89.6–873.1 GHz (Pickett et al. 1998), a median 0.9 GHz apart. No crossing of either sample falls within 25 MHz of one; the median separation is 0.81 GHz; and 1 crossing lies inside the same $\pm 50 \text{ km s}^{-1}$ window used above, against 2.1 expected from the windows’ own channel grids. The one concentration a reader will notice — 15 crossings of the census and the hold-out between 344.2 and 346.4 GHz — is not atmospheric either, because that interval holds 0 telluric transitions; 12 of the 43 unattributed crossings whose window the catalogue tabulates fall in it against 11.8 ± 0.9 expected from where the spectral windows were tuned. *It is the carbon monoxide tuning, seen from the other side.*

Third, fixed intermediate frequency. An artefact of the signal chain would sit at a fixed position within the spectral window rather than at a fixed sky frequency. This test is run over every window for which the catalogue releases a peak frequency — 1397 of the 1651, of which only 51 carry a threshold crossing — because such an artefact would occupy the same intermediate frequency in all of them and not only where the trigger fires. It does not: the peaks’ fractional positions within their own windows have median 0.50 and are uniform.

Fourth, a cross-target occupancy screen over every window the pipeline extracted, and not only the crossings. That set is wider than the census: 3027 windows toward 133 stars, being the census together with the rows its audit removed, the reserved hold-out, the archival extension, the out-of-sample control, and 64 windows in 16

blocks that entered none of them. Interference belongs to the observatory and not to the star, so a terrestrial carrier should recur at the same sky frequency toward *unrelated* stars. It does not even for the leading event: the one unattributed rank-passing event has a measured crossing frequency; 40 further windows, toward 24 other stars in 32 other blocks, cover that frequency, and none of them carries a crossing in the same channel. With each window’s peak randomised on its own channel grid, the number of cross-star coincidences within 5 MHz is 569 against 554 ± 13 expected ($p = 0.15$); among the 109 windows whose peak exceeds the trigger it is 4 against 5.2 ± 1.9 , i.e. *below* the null, and with the molecular-line mask applied 1 against 2.1 ± 1.3 . There is no excess to explain. The mask has to be applied first, because 35 of those 109 windows sit on a masked transition (CO(2→1) $\times 17$, CO(1→0) $\times 10$, CO(3→2) $\times 4$, $^{13}\text{CO}(2\rightarrow 1) \times 3$, SiO(5→4) $\times 1$), and a carbon monoxide line falls at nearly the same sky frequency toward every nearby star — exactly the coincidence the screen is looking for.

One execution block fails these tests, and it fails them as a block rather than as a carrier. Its 4 crossings toward BD+05 1668 are the highest statistics in the survey outside β Pictoris and HD 48370, and the field statistics that identify the failure are in §5.3.2. What belongs here is whether interference accounts for it, and three readings say not. The visibility fit returns a large imaginary part, placing the emission away from the phase centre. The block’s 24 sibling executions of the same field, with the same array and the same spectral setup, produce no crossing at all. And it is not a carrier at a fixed sky frequency: each of the four frequencies is covered by 127–137 further windows toward 17–22 other stars, and 0 of them carries a crossing in the same channel.

One reading does point at a mechanism, and it can be tested across the survey. 2 of the 4 sit at the identical channel of two different spectral windows, which is the signature of a fixed intermediate frequency — and a feature with no geometric fringe phase, a spur common to every antenna being the obvious case, does not fringe-track, so it gathers at the phase centre rather than at the star. Here the two are a twentieth of an arcsecond apart, so the block cannot test its own hypothesis; the survey can, because the star lies 5.0–8.1 arcsec from the phase centre in 75 windows whose control annulus brackets it, putting a median of 15 probes within two synthesised beams of the phase centre and the star well outside. Against the probes at the same distance from the star but elsewhere in azimuth — matching the radius matters, or a centrally concentrated disc would imitate the effect — those probes are higher by $+0.09 \pm 0.08$ in the statistic, consistent with zero and bounded by 0.26, smaller by a factor of about 33 than the 8.4 by which this block’s ring is raised. *The phase centre is not where the excess lives.* The evidence supports a data-quality failure of one execution block, and the block criterion of Appendix D.2 — which fails 1 block of the 404 searched — is what disposes of these 4 crossings (§5.3.2).

None of this is a substitute for the ON/OFF cadence a dedicated search uses, and we do not present it as one. We claim only that *no identified interference mechanism accounts for the unattributed events*: allocation tables cannot exclude harmonics, intermodulation prod-

ucts, unconventional emitters or near-field sources, and archival data without a simultaneous off-source strategy cannot do better. The four tests establish that the cross-

ing population as a whole carries none of the signatures interference leaves, which is a statement about the population and not a clearance of any individual crossing.

REFERENCES

- Akeson R. L., et al., 2013, *PASP*, 125, 989
 Astropy Collab., 2022, *ApJ*, 935, 167
 Backus P. R., Project Phoenix Team, 2002, in Stanimirović S., Altschuler D., Goldsmith P., Salter C., eds, *ASP Conf. Ser. Vol. 278, Single-Dish Radio Astronomy: Techniques and Applications*. Astron. Soc. Pac., San Francisco, p. 525
 Burton K., MacGregor M. A., Osten R. A., 2022, *ApJL*, 939, L6
 CASA Team, 2022, *PASP*, 134, 114501
 Cataldi G., et al., 2023, *ApJ*, 951, 111
 Choza C., et al., 2024, *AJ*, 167, 10
 Cocconi G., Morrison P., 1959, *Nature*, 184, 844
 Cortes P. C., et al., 2026, *ALMA Cycle 13 Technical Handbook*, ALMA Doc. 13.3, ver. 1.0, doi:10.5281/zenodo.18793803
 Czech D. J., et al., 2021, *PASP*, 133, 064502
 Dame T. M., Hartmann D., Thaddeus P., 2001, *ApJ*, 547, 792
 Dame T. M., Thaddeus P., 2022, *ApJS*, 262, 5
 Dent W. R. F., et al., 2014, *Science*, 343, 1490
 Drake F. D., 1961, *Physics Today*, 14(4), 40
 Enriquez J. E., et al., 2017, *ApJ*, 849, 104
 Gaia Collab., Vallenari A., et al., 2023, *A&A*, 674, A1
 Garrett M. A., 2026, *MNRAS*, in press (arXiv:2605.10212)
 Ginsburg A., et al., 2019, *AJ*, 157, 98
 Harris C. R., et al., 2020, *Nature*, 585, 357
 Hunter T. R., Phillips N. M., Brogière D., González J., 2018, *Improving the Calibration of Atmospheric Spectral Features in ALMA Data*. NA ALMA 2016 Development Study Final Report, 2018 March 2
 International Telecommunication Union, 2024, *Radio Regulations*, Edition of 2024. ITU, Geneva
 Jacobson-Bell B., et al., 2025, *AJ*, 169, 206
 Johnson O. A., et al., 2023, *AJ*, 166, 193
 Kamenno S., 2014, *ALMA Cross Correlation Bandpass Stability*. ALMA Technical Note 15 (draft), 2014 December 17
 Li J.-K., Tao Z.-Z., Zhang T.-J., 2026, *ApJ*, 1006, 163
 Ma P. X., et al., 2023, *NatAs*, 7, 492
 MacGregor M. A., et al., 2020, *ApJ*, 891, 80
 Manunza L., et al., 2025, *AcAau*, 233, 155
 Margot J.-L., et al., 2023, *AJ*, 166, 206
 Mason L. A., Garrett M. A., Wandia K., Siemion A. P. V., 2025, *MNRAS*, 536, 2127
 Mauersberger R., Wilson T. L., Rood R. T., Bania T. M., Hein H., Linhart A., 1996, *A&A*, 306, 141
 Matrà L., et al., 2017, *MNRAS*, 464, 1415
 Morrison I. S., et al., 2023, *PASA*, 40, e019
 Müller H. S. P., et al., 2005, *JMoSt*, 742, 215
 Pardo S., et al., 2025, *AJ*, 170, 12
 Pickett H. M., et al., 1998, *JQSRT*, 60, 883
 Planck Collab., Adam R., et al., 2016, *A&A*, 594, A10
 Price D. C., et al., 2020, *AJ*, 159, 86
 Privon G., et al., 2026, *ALMA Cycle 13 Proposer's Guide*, ALMA Doc. 13.2, ver. 1.0, doi:10.5281/zenodo.18793859
 Remijan A. J., Markwick-Kemper A., ALMA Working Group on Spectral Line Frequencies, 2007, in *American Astronomical Society Meeting Abstracts* 211, 132.11
 Richards A. M. S., et al., 2022, *Self-calibration and improving image fidelity for ALMA and other radio interferometers*, ALMA Memo 620
 Sheikh S. Z., Wright J. T., Siemion A. P. V., Enriquez J. E., 2019, *ApJ*, 884, 14
 Sheikh S. Z., 2020, *IJAsB*, 19, 237
 Sheikh S. Z., et al., 2021, *Nature Astronomy*, 5, 1153
 Sheikh S. Z., et al., 2025, *AJ*, 169, 118
 Steffes P. G., DeBoer D. R., 1994, *Icarus*, 107, 215
 Stephens G. L., et al., 2002, *Bull. Am. Meteorol. Soc.*, 83, 1771
 Tarter J., 2001, *ARA&A*, 39, 511
 Tremblay C. D., Tingay S. J., 2024, *ApJ*, 972, 76
 Vidal C., et al., 2026, arXiv:2605.21093 (preprint)
 Virtanen P., et al., 2020, *NatMe*, 17, 261
 Wehr T., et al., 2023, *Atmos. Meas. Tech.*, 16, 3581
 Wenger M., et al., 2000, *A&AS*, 143, 9
 Wright J. T., et al., 2018, *AJ*, 156, 260
 Zhang Z.-S., et al., 2020, *ApJ*, 891, 174